

TARGET-90

JKBOSE CLASS-11TH

Classification of Elements and
Periodicity in Properties

One Shot

CHEMISTRY

Vishal Sir



Bhaichara
ON
TOP



JKBose Wallah

Topics To Be Covered

1 ... One Shot

2 ...

3 ...

4 ...

➤ **UNIT-III: Classification of Element and Periodicity in Properties**

- Significance of classification, brief history of the development of periodic table. Modern periodic law and the present form of the periodic table, periodic trends in properties of elements: atomic radii, ionic radii, inert gas radii, ionization enthalpy, electron gain enthalpy, electronegativity, valency.

Need for Classification of Elements

- At the end of the 18th century, only 31 elements were known, making it easy to remember their names and properties. With time, more elements were discovered, and today the total has reached 118. This made it difficult to memorize the properties of all elements.
- To overcome this difficulty, classification of elements became necessary. Classification is the systematic arrangement of elements in such a way that those with similar properties are grouped together, while those with different properties are kept separate.

Brief History of Classification of Elements

1. DOBEREINER'S TRIADS:

- A German scientist, Johann Dobereiner, grouped elements in sets of three called triads.
- Elements in a triad had similar properties.
- The atomic mass of the middle element was nearly equal to the average of the other two.

Examples:

- Lithium, Sodium, Potassium
- Calcium, Strontium, Barium
- Chlorine, Bromine, Iodine

Element	Atomic weight	
Lithium Sodium Potassium	7 23 39	Mean of 1st and 3rd $\frac{7 + 39}{2} = 23$
Calcium Strontium Barium	40 88 137	Mean of 1st and 3rd $\frac{40 + 137}{2} = 88.5$
Chlorine Bromine Iodine	35.5 80 127	Mean of 1st and 3rd $\frac{35.5 + 127}{2} = 81.25$

Brief History of Classification of Elements

2. NEWLANDS LAW OF OCTAVES:

- In 1865, John Newlands, an English chemist, suggested a new way of arranging elements. He grouped elements by increasing atomic weight and noticed that every eighth element showed properties similar to the first one.
- He compared this to musical octaves, where every eighth note is similar to the first. However, this law worked only for lighter elements up to calcium and failed for heavier ones.

Element	Li	Be	B	C	N	O	F
At. wt.	7	9	11	12	14	16	19
Element	Na	Mg	Al	Si	P	S	Cl
At. wt.	23	24	27	29	31	32	35.5
Element	K	Ca					
At. wt.	39	40					

~~X~~



Li

Be

B

C

N

O

F

~~X~~

Na

Mg

Al

Si

P

S

Cl

~~X~~

K

Ca

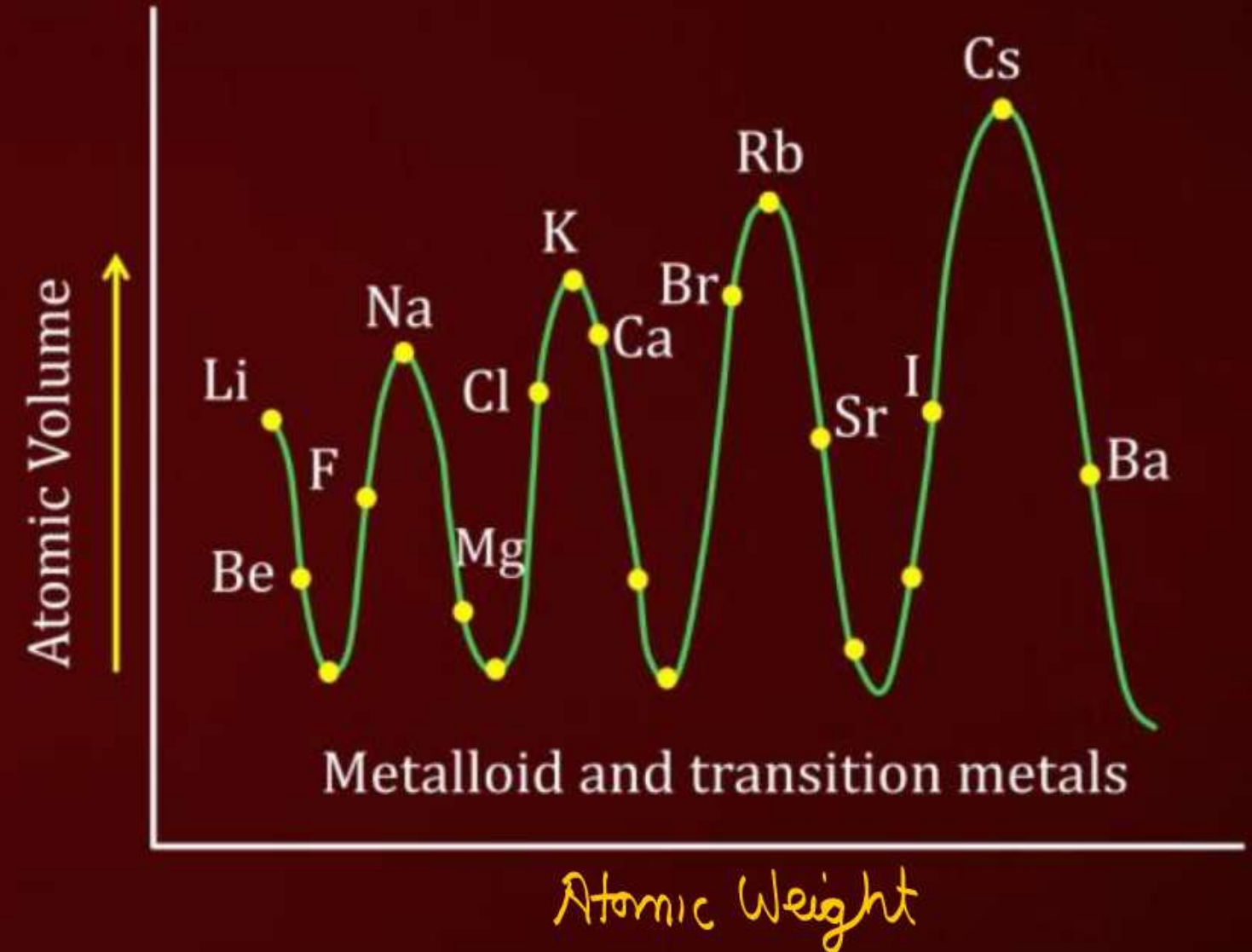
Brief History of Classification of Elements

3. LOTHER MEYER'S ARRANGEMENT:

- Lotther Meyer, a German chemist, studied properties like atomic volume, melting point, and boiling point of elements.
- He plotted these properties against atomic weights and found that elements with similar properties appeared at similar positions on the graph.

Examples:

- Strongly electropositive alkali metals (Li, Na, K, Rb, Cs) → at the peaks of the curve.
- Less electropositive alkaline earth metals (Be, Mg, Ca, Sr, Ba) → at the descending parts.
- Highly electronegative halogens (F, Cl, Br, I) → at the ascending parts.



Brief History of Classification of Elements

Mendeleev's Periodic Law and Table

- In 1869, Mendeleev arranged elements in order of increasing atomic weights and grouped them with similar properties. (Mendeleev's Periodic Law)
- He proposed Periodic Law: Physical and chemical properties of elements are periodic functions of their atomic weights.
- He left gaps for undiscovered elements and predicted their properties correctly.
- For example, he predicted eka-aluminium (later discovered as gallium) and eka-silicon (later discovered as germanium).
- The discovery of these elements proved the accuracy of his periodic table.

Characteristics of Mendeleev's Periodic Table

- **Mendeleev's periodic table has:**
 - Eight vertical columns called groups (I to VIII). Except group VIII, each group is divided into two sub-groups A and B. A separate group zero was later added for inert gases.
 - Six horizontal rows called periods or series, numbered from 1 to 6.

Groups	I		II		III		IV		V		VI		VII		VIII			
Oxides Hydrides	RO RH		BO RH ₂		R ₂ O ₃ RH ₃		RO ₂ RH ₄		R ₂ O ₅ RH ₃		RO ₃ RH ₂		R ₂ O ₇ RH		RO ₄			
Periods ↓	A	B	A	B	A	B	A	B	A	B	A	B	A	B	Transition series			
1	H 1.808																	
2	Li 6.939		Be 6.939		B 6.939		C 12.011		N 14.007		O 15.999		F 18.998					
3	Na 22.99		Mg 24.31		Al 29.98		Si 28.09		P 30.974		S 32.06		Cl 35.455					
4 First series Second series	K 39.102 Cu 63.54		Ca 40.08 Zn 65.37		Sc 44.96 Ga 69.72		Ti 47.90 Ge 72.59		V 50.94 As 74.92		Cr 50.20 Se 78.96		Mn 54.94 Br 79.909		Fe 55.85	Co 58.93	Ni 58.71	
5 First series Second series	Rb 85.47 Ag 107.87		Sr 87.62 Cd 112.40		Y 88.91 Id 114.82		Zr 91.22 Sn 118.69		Nb 92.91 Sb 121.75		Mo 95.94 Te 127.60		Tc 99 I 126.90		Ru 101.07	Rh 102.91	Pd 106.4	
6 First series Second series	Cs 132.90 Au 196.97		Ba 137.34 Hg 200.59		La 138.91 Tl 204.57		Hf 178.49 Pb 207.19		Ta 180.95 Bi 208.98		W 183.85				Os 190.2	Ir 192.2	Pt 135.09	

Significance of Mendeleev's Periodic Table

- 1. Systematic Study:** Grouping made learning and predicting properties easier.
- 2. Correction of Atomic Masses:** Wrong atomic masses (Be, In, Au) were corrected.
- 3. Prediction of New Elements:** Left gaps and predicted elements (eka-Al → Gallium, eka-Si → Germanium) which were later discovered.

Defects in of Mendeleev's Periodic Table

- 1. Position of Hydrogen:** Properties of Hydrogen match both alkali metals and halogens, so placement was uncertain.
- 2. Anomalous Pairs:** Some elements ^{were} not in atomic mass order (increasing) (e.g., Argon-Potassium, Cobalt-Nickel).
- 3. Isotopes:** Elements with same atomic number but different masses had no separate position.
- 4. Cause of Periodicity:** Mendeleev could not explain why properties repeat.
- 5. f-block Elements:** Lanthanoids and Actinoids were not given proper positions.

Modern Periodic Law



- In 1913, Moseley studied the X-rays emitted by elements when bombarded with high-speed electrons. He concluded that the square root of X-ray frequency ($\sqrt{\nu}$) is directly proportional to the atomic number (Z) of the element:

$$\sqrt{\nu} \propto Z$$

- He suggested that atomic number (Z), not atomic weight, is the fundamental property of elements.
- "The physical and chemical properties of elements are the periodic function of their atomic numbers."

Modern Periodic Table

- Also known as Bohr's Table
- Characteristics.
 - It consists of 7 horizontal rows called as Periods (1 to 7).
 - And 18 vertical columns called as Groups (1 to 18).

PERIODIC TABLE OF ELEMENTS



Group →
Period ↓ 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H Hydrogen Nonmetal																	2 He Helium Noble Gas
3 Li Lithium Alkali Metal	4 Be Beryllium Alkaline Earth Metal											5 B Boron Metalloid	6 C Carbon Nonmetal	7 N Nitrogen Nonmetal	8 O Oxygen Nonmetal	9 F Fluorine Halogen	10 Ne Neon Noble Gas
11 Na Sodium Alkali Metal	12 Mg Magnesium Alkaline Earth Metal											13 Al Aluminium Post-Transition M...	14 Si Silicon Metalloid	15 P Phosphorus Nonmetal	16 S Sulfur Nonmetal	17 Cl Chlorine Halogen	18 Ar Argon Noble Gas
19 K Potassium Alkali Metal	20 Ca Calcium Alkaline Earth Metal	21 Sc Scandium Transition Metal	22 Ti Titanium Transition Metal	23 V Vanadium Transition Metal	24 Cr Chromium Transition Metal	25 Mn Manganese Transition Metal	26 Fe Iron Transition Metal	27 Co Cobalt Transition Metal	28 Ni Nickel Transition Metal	29 Cu Copper Transition Metal	30 Zn Zinc Transition Metal	31 Ga Gallium Post-Transition M...	32 Ge Germanium Metalloid	33 As Arsenic Metalloid	34 Se Selenium Nonmetal	35 Br Bromine Halogen	36 Kr Krypton Noble Gas
37 Rb Rubidium Alkali Metal	38 Sr Strontium Alkaline Earth Metal	39 Y Yttrium Transition Metal	40 Zr Zirconium Transition Metal	41 Nb Niobium Transition Metal	42 Mo Molybdenum Transition Metal	43 Tc Technetium Transition Metal	44 Ru Ruthenium Transition Metal	45 Rh Rhodium Transition Metal	46 Pd Palladium Transition Metal	47 Ag Silver Transition Metal	48 Cd Cadmium Transition Metal	49 In Indium Post-Transition M...	50 Sn Tin Post-Transition M...	51 Sb Antimony Metalloid	52 Te Tellurium Metalloid	53 I Iodine Halogen	54 Xe Xenon Noble Gas
55 Cs Cesium Alkali Metal	56 Ba Barium Alkaline Earth Metal	57-71 ↓	72 Hf Hafnium Transition Metal	73 Ta Tantalum Transition Metal	74 W Tungsten Transition Metal	75 Re Rhenium Transition Metal	76 Os Osmium Transition Metal	77 Ir Iridium Transition Metal	78 Pt Platinum Transition Metal	79 Au Gold Transition Metal	80 Hg Mercury Transition Metal	81 Tl Thallium Post-Transition M...	82 Pb Lead Post-Transition M...	83 Bi Bismuth Post-Transition M...	84 Po Polonium Metalloid	85 At Astatine Halogen	86 Rn Radon Noble Gas
87 Fr Francium Alkali Metal	88 Ra Radium Alkaline Earth Metal	89-103 ↓	104 Rf Rutherfordium Transition Metal	105 Db Dubnium Transition Metal	106 Sg Seaborgium Transition Metal	107 Bh Bohrium Transition Metal	108 Hs Hassium Transition Metal	109 Mt Meitnerium Transition Metal	110 Ds Darmstadtium Transition Metal	111 Rg Roentgenium Transition Metal	112 Cn Copernicium Transition Metal	113 Nh Nihonium Post-Transition M...	114 Fl Flerovium Post-Transition M...	115 Mc Moscovium Post-Transition M...	116 Lv Livermorium Post-Transition M...	117 Ts Tennessine Halogen	118 Og Oganesson Noble Gas
Lanthanides (57 – 71)		57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium	
Actinides (89 – 103)		89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium	

Atomic Number → 17

Atomic Mass, u → 35.45

Symbol → Cl

Name → Chlorine

Chemical Group Block → Halogen

- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Post-Transition Metals
- Metalloids
- Nonmetals
- Halogens
- Noble Gases
- Lanthanides
- Actinides

PERIODIC TABLE OF ELEMENTS



Group →
Period ↓

1
2
3
4
5
6
7

Atomic Number → 17 35.45 ← Atomic Mass, u

Symbol → Cl

Name → Chlorine Halogen

Chemical Group Block

1 H Hydrogen Nonmetal	2 He Helium Noble Gas																
3 Li Lithium Alkali Metal	4 Be Beryllium Alkaline Earth Metal	5 B Boron Metalloid	6 C Carbon Nonmetal	7 N Nitrogen Nonmetal	8 O Oxygen Nonmetal	9 F Fluorine Halogen	10 Ne Neon Noble Gas										
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Atomic Number

17

Atomic Mass, u

35.45

Symbol

Cl

Name

Chlorine

Chemical Group Block

Halogen

Lanthanides (57-71)	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
Actinides (89-103)	89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Post-Transition Metals
- Metalloids
- Nonmetals
- Halogens
- Noble Gases
- Lanthanides
- Actinides

Electronic Configuration in Periods

Period	No. of Elements	Subshells Involved	Element Range
1st	2	1s	H(1) – He(2)
2nd	8	2s, 2p	Li(3) – Ne(10)
3rd	8	3s, 3p	Na(11) – Ar(18)
4th	18	4s, 3d, 4p	K(19) – Kr(36)
5th	18	5s, 4d, 5p	Rb(37) – Xe(54)
6th	32	6s, 4f, 5d, 6p	Fr(87) – Og(118)
7th	32	7s, 5f, 6d, 7p	Fr(87) – Og(118)

2 → s
 8 → sp
 18 → s d p
 32 → s f d p

Modern Periodic Table

BLOCKS → Four blocks (s, p, d & f).

2nd (2s ¹)	Li	Be → 2s ²
3rd (3s ¹)	Na	Mg
(4s ¹)	K	Ca
(5s ¹)	Rb	Sr → 5s ²
(6s ¹)	Cs	Ba
(7s ¹)	Fr	Ra → 7s ²

1) S-Block → Elements in which last enters in s-orbital of valence shell.

General electronic configuration: ns^{1-2} .

Consist of two Groups

i) Group 1 (ns^1) → Also called as Alkali metals (H, Li, Na, K, Rb, Cs, Fr).

ii) Group 2 (ns^2) → Also called as Alkaline earth metals. (Be, Mg, Ca, Sr, Ba, Ra).

Group 1 (ns^1):

✓	Period 1:	$1s^1$	→	H
✓	Period 2:	$2s^1$	→	Li
✓	Period 3:	$3s^1$	→	Na
✓	Period 4:	$4s^1$	→	K
✓	Period 5:	$5s^1$	→	Rb
✓	Period 6:	$6s^1$	→	Cs
✓	Period 7:	$7s^1$	→	Fr

Group 2 (ns^2):

✓	Period 2:	$2s^2$	→	Be
✓	Period 3:	$3s^2$	→	Mg
✓	Period 4:	$4s^2$	→	Ca
✓	Period 5:	$5s^2$	→	Sr
✓	Period 6:	$6s^2$	→	Ba
✓	Period 7:	$7s^2$	→	Ra



Electronic Configuration of s-Block Elements

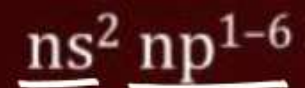


Division of Elements into s, p, d and f-Block

p-block elements → Elements in which the last electron enters in p-orbital of outermost shell.



General Electronic Configuration:



It involves groups from 13 to 18

- **Group 13** elements are called Boron family & has $ns^2 np^1$ configuration.
- **Group 14** → Carbon Family & has $ns^2 np^2$
- **Group 15** → Pnictogens & has $ns^2 np^3$
- **Group 16** → Chalcogens & has $ns^2 np^4$
- **Group 17** → Halogens & has $ns^2 np^5$ electronic config.

18 → Nobel gases & has $ns^2 np^6$ except He ($1s^2$).

13	14	15	16	17	18
$ns^2 np^1$	$ns^2 np^2$	$ns^2 np^3$		$ns^2 np^5$	He
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	Rn
Nh	Fl	Mc	Lv	Ts	Og

PERIODIC TABLE OF ELEMENTS



Group →
Period ↓ 1

1

2

3

4

5

6

7

1 1.0080 H Hydrogen Nonmetal	2	3 17 35.45 Cl Chlorine Halogen	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18 2 4.00260 He Helium Noble Gas
3 6.94 Li Lithium Alkali Metal	4 9.012183 Be Beryllium Alkaline Earth Metal											5 10.81 B Boron Metalloid	6 12.011 C Carbon Nonmetal	7 14.007 N Nitrogen Nonmetal	8 15.999 O Oxygen Nonmetal	9 18.9984 F Fluorine Halogen	10 20.180 Ne Neon Noble Gas
11 22.989... Na Sodium Alkali Metal	12 24.305 Mg Magnesium Alkaline Earth Metal											13 26.981... Al Aluminium Post-Transition M...	14 28.085 Si Silicon Metalloid	15 30.973... P Phosphorus Nonmetal	16 32.07 S Sulfur Nonmetal	17 35.45 Cl Chlorine Halogen	18 39.9 Ar Argon Noble Gas
19 39.0983 K Potassium Alkali Metal	20 40.08 Ca Calcium Alkaline Earth Metal	21 44.9559 Sc Scandium Transition Metal	22 47.867 Ti Titanium Transition Metal	23 50.9415 V Vanadium Transition Metal	24 51.996 Cr Chromium Transition Metal	25 54.9380 Mn Manganese Transition Metal	26 55.84 Fe Iron Transition Metal	27 58.93319 Co Cobalt Transition Metal	28 58.693 Ni Nickel Transition Metal	29 63.55 Cu Copper Transition Metal	30 65.4 Zn Zinc Transition Metal	31 69.723 Ga Gallium Post-Transition M...	32 72.63 Ge Germanium Metalloid	33 74.92159 As Arsenic Metalloid	34 78.97 Se Selenium Nonmetal	35 79.90 Br Bromine Halogen	36 83.80 Kr Krypton Noble Gas
37 85.468 Rb Rubidium Alkali Metal	38 87.62 Sr Strontium Alkaline Earth Metal	39 88.90584 Y Yttrium Transition Metal	40 91.22 Zr Zirconium Transition Metal	41 92.90637 Nb Niobium Transition Metal	42 95.95 Mo Molybdenum Transition Metal	43 98.90636 Tc Technetium Transition Metal	44 101.1 Ru Ruthenium Transition Metal	45 102.9055 Rh Rhodium Transition Metal	46 106.42 Pd Palladium Transition Metal	47 107.868 Ag Silver Transition Metal	48 112.41 Cd Cadmium Transition Metal	49 114.818 In Indium Post-Transition M...	50 118.71 Sn Tin Post-Transition M...	51 121.760 Sb Antimony Metalloid	52 127.6 Te Tellurium Metalloid	53 126.9045 I Iodine Halogen	54 131.29 Xe Xenon Noble Gas
55 132.90... Cs Cesium Alkali Metal	56 137.33 Ba Barium Alkaline Earth Metal	57-71 Lanthanides	72 178.49 Hf Hafnium Transition Metal	73 180.9479 Ta Tantalum Transition Metal	74 183.84 W Tungsten Transition Metal	75 186.207 Re Rhenium Transition Metal	76 190.2 Os Osmium Transition Metal	77 192.22 Ir Iridium Transition Metal	78 195.08 Pt Platinum Transition Metal	79 196.96... Au Gold Transition Metal	80 200.59 Hg Mercury Transition Metal	81 204.383 Tl Thallium Post-Transition M...	82 207 Pb Lead Post-Transition M...	83 208.98... Bi Bismuth Post-Transition M...	84 208.98... Po Polonium Metalloid	85 209.98... At Astatine Halogen	86 222.01... Rn Radon Noble Gas
87 223.01... Fr Francium Alkali Metal	88 226.02... Ra Radium Alkaline Earth Metal	89-103 Actinides	104 267.1... Rf Rutherfordium Transition Metal	105 268.1... Db Dubnium Transition Metal	106 269.1... Sg Seaborgium Transition Metal	107 270.1... Bh Bohrium Transition Metal	108 269.1... Hs Hassium Transition Metal	109 277.1... Mt Meitnerium Transition Metal	110 282.1... Ds Darmstadtium Transition Metal	111 282.1... Rg Roentgenium Transition Metal	112 286.1... Cn Copernicium Transition Metal	113 286.1... Nh Nihonium Post-Transition M...	114 290.1... Fl Flerovium Post-Transition M...	115 290.1... Mc Moscovium Post-Transition M...	116 293.2... Lv Livermorium Post-Transition M...	117 294.2... Ts Tennessine Halogen	118 295.2... Og Oganesson Noble Gas

Handwritten notes in Thai script:
 1. Highest 'n' (referring to group 1)
 2. Atomic Number (referring to Cl)
 3. Atomic Mass, u (referring to Cl)
 4. Symbol (referring to Cl)
 5. Name (referring to Cl)
 6. Chemical Group Block (referring to Cl)
 7. n, n, n, n, n (referring to groups 13-17)

Handwritten notes in Thai script:
 1. Lanthanides (57-71)
 2. Actinides (89-103)

Lanthanides (57 – 71)	57 138.905 Ce Cerium	58 140.116 Pr Praseodymium	59 140.9076 Nd Neodymium	60 144.24 Pm Promethium	61 144.9127 Sm Samarium	62 150.4 Eu Europium	63 151.964 Gd Gadolinium	64 157.2 Tb Terbium	65 158.9253 Dy Dysprosium	66 162.500 Ho Holmium	67 164.9303 Er Erbium	68 167.26 Tm Thulium	69 168.9342 Yb Ytterbium	70 173.05 Lu Lutetium
Actinides (89 – 103)	89 227.03 Th Thorium	90 232.038 Pa Protactinium	91 231.03... U Uranium	92 238.0289 Np Neptunium	93 237.04... Pu Plutonium	94 244.06... Am Americium	95 243.06... Cm Curium	96 247.07... Bk Berkelium	97 247.07... Cf Californium	98 251.08 Es Einsteinium	99 252.0830 Fm Fermium	100 257.0... Md Mendelevium	101 258.0... No Nobelium	102 259.1... Lr Lawrencium

- Alkali Metals
- Alkaline Earth Metals
- Transition Metals
- Post-Transition Metals
- Metalloids
- Nonmetals
- Halogens
- Noble Gases
- Lanthanides
- Actinides

Division of Elements into s, p, d and f-Block

d-Block Elements

- Elements where the last electron enters the d-orbital of the (n-1) shell (penultimate shell). ⁱⁿ
- General Electronic Configuration: $(n-1)d^{1-10} ns^{0-2}$
- Groups: 3 to 12
- Includes four series of elements.
- Also called Transition Elements because they show a transition in properties from electropositive to electronegative character.
- **1st Series (3d-series):**
4th period → Sc (Z = 21) to Zn (Z = 30)
- **2nd Series (4d-series):**
5th period → Y (Z = 39) to Cd (Z = 48)
- **3rd Series (5d-series):**
6th period → La (Z = 57) to Hg (Z = 80)
- **4th Series (6d-series):**
7th period → Ac (Z = 89) to Cn (Z = 112)

4th " " (6d) : 7th Period

[illegible]

Division of Elements into s, p, d and f-Block

f-Block Elements

- Last electron enters the f -orbital of the $(n-2)$ shell.
- General configuration: $(n-2) f^{1-14} (n-1) d^{0-1} ns^2$
- All f -block elements belong to Group 3.
- They are called inner-transition elements.
- There are two series of 14 elements, each.

1. Lanthanides (4f-series):

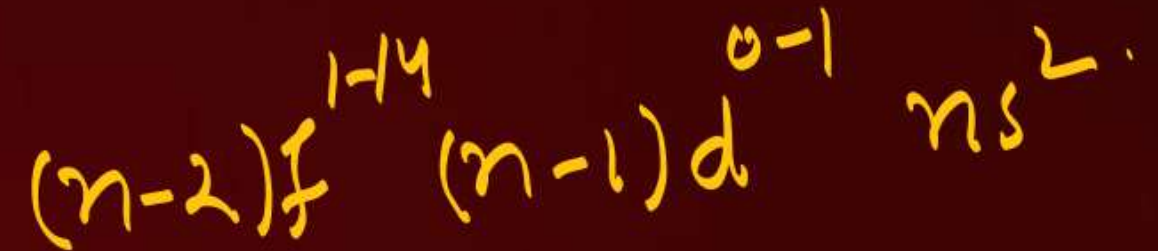
Present in 6th period

From Ce ($Z = 58$) to Lu ($Z = 71$)

2. Actinides (5f-series):

Present in 7th period

From Th ($Z = 90$) to Lr ($Z = 103$)



Division of Elements into s, p, d and f-Block

La ✓

58 ✓	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu 71 ✓
90 ✓	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Em	Md	No	Lr 103 ✓

✓ Planets ✓ Places ✓ Scientist

Nomenclature of Elements having Atomic Number more than hundred



1 0 1

Unnil unium Unu

1 0 2

Unnil bi um Unb.

1 0 3

Unnil tri um. Unt.

end
tium

Digit	Name	Abbreviation
0	nil	<i>n</i>
1	un	<i>u</i>
2	bi	<i>b</i>
3	tri	<i>t</i>
4	quad	<i>q</i>
5	pent	<i>p</i>
6	hex	<i>h</i>
7	sept	<i>s</i>
8	oct	<i>o</i>
9	enn	<i>e</i>

Atomic Number	Name according to IUPAC nomenclature	Symbol	IUPAC Official Name	IUPAC Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Ununnilium	Uun	Darmstadtium	Ds
111	Unununnium	Uuu	Rontgenium	Rg
112	Ununbium	Uub	Copernicium	Cn
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

→
 →
 →
 →
 →
 → Saage
 → Bhari
 → Hasi
 → Mister
 → Danish
 → Real god
 → Can
 → Never
 → Feel
 → Much
 → Love for
 → These
 → organisms.

Nomenclature of Elements having Atomic Number more than hundred



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101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
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106	Unnilhexium	Unh	Seaborgium	Sg
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108	Unniloctium	Uno	Hassium	Hs
109	Unnilenium	Une	Meitnerium	Mt
110	Ununnillium	Uun	Darmstadtium	Ds

Nomenclature of Elements having Atomic Number more than hundred



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116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

PERIODIC PROPERTIES.

Radius

IE

ΔG°

Electro-
negativity

Valency

Diagonal
Relationship

Atomic Radius



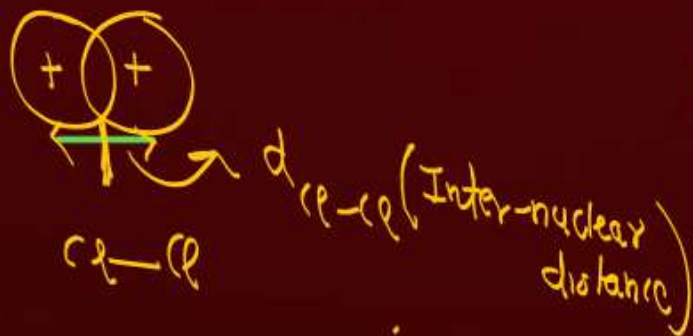
- Distance from the nucleus to the outermost electron shell.
- Measured in Å (Angstrom) or pm (Picometer).
- Atoms have no sharp boundary, so radius is measured in a bonded state.

Types of Atomic Radius:

1. Covalent Radius (for non-metals)

- Half the distance between nuclei of two same atoms bonded by a covalent bond.

$$\text{Formula: } r_{\text{cov}} = \frac{d_{\text{A-A}}}{2}$$



2. Metallic Radius (for metals)

- Half the distance between nuclei of two adjacent atoms in a metallic lattice.

$$\text{Formula: } r_{\text{metallic}} = \frac{d_{\text{M-M}}}{2}$$



3. Van der Waals Radius

(for inert gases/non-bonded atoms)

- Half the distance between nuclei of two same non-bonded atoms in different molecules.

$$\text{Formula: } r_{\text{van}} = \frac{d_{\text{A-A}}}{2}$$



$$r_{\text{cov}} < r_{\text{metallic}} < r_{\text{van}}$$

Atomic Radius



Screening/Shielding Effect

- Inner shell (core) electrons block the outermost electrons from the full pull of the nucleus.
- This is called shielding (screening) effect.

Effective Nuclear Charge (Z^*)

- Net attraction felt by valence electrons after shielding.
- Higher Z^* → stronger pull by nucleus → smaller atomic size.

Variation of Atomic Radius



1. Across a Period (→)

- Atomic number ↑, Proton number ↑
- Effective nuclear charge ↑
- Electrons enter the same shell
- Attraction ↑ → Size decreases

Example (2nd period):

Li > Be > B > C > N > O > F

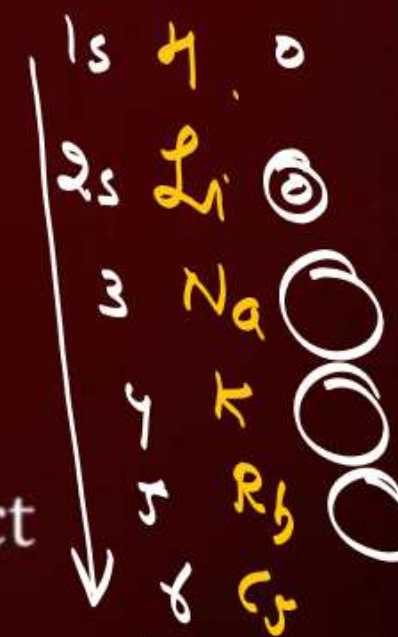
Le van
Ne

2. Down a Group (↓)

- Atomic number ↑
- New shells added → size ↑
- Effect of new shells > Z^* effect

Example (alkali metals):

Li < Na < K < Rb < Cs



Ionic Radius

- Defined only for **ionic compounds**.
- Distance from nucleus to the point of ion's influence
- Formula:** $d = r^+ + r^-$
 - r^+ = radius of cation
 - r^- = radius of anion
- Measured in Å/pm (using X-ray spectroscopy).



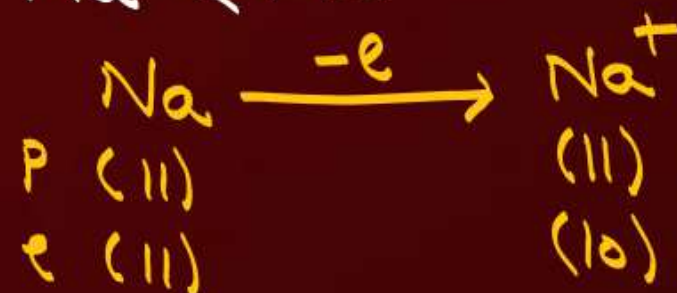
Trends:

- Cation radius < parent atom ($\text{Na}^+ < \text{Na}$)
→ fewer electrons, higher Z^* , stronger attraction.
- Anion radius > parent atom ($\text{F}^- > \text{F}$)
more electrons, lower Z^* , weaker attraction.

Isoelectronic Species

- Species with same number of electrons.
- Example:** O^{2-} , F^- , Na^+ , Mg^{2+} , Al^{3+} (all have $10e^-$).
- Rule:** Ionic radius $\propto 1/\text{Nuclear charge}$.
- Order:** $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

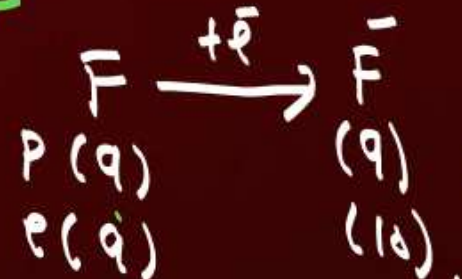
eg $\text{Na}^+ < \text{Na}$



i.e. $Z_{\text{eff}} \uparrow$ size $\downarrow$

$\text{Na}^+ < \text{Na}$

eg $\text{F}^- > \text{F}$



$Z_{\text{eff}} \downarrow$ size $\uparrow$

$\text{F}^- > \text{F}$

$\bar{A} > A > A^+$

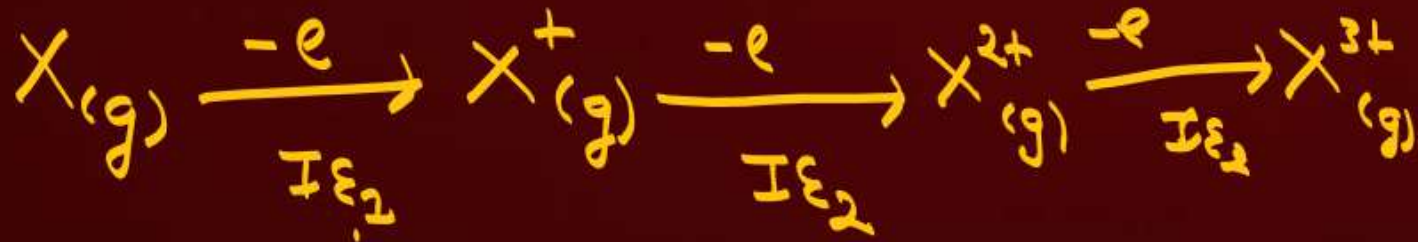
Ionization Enthalpy (IE)

- Minimum energy required to remove an electron from an isolated gaseous atom.
- Units: kJ mol^{-1} or eV/atom .

$$X_{(g)} \xrightarrow{-e} X^+ + \text{IE}$$

First & Successive IE: order

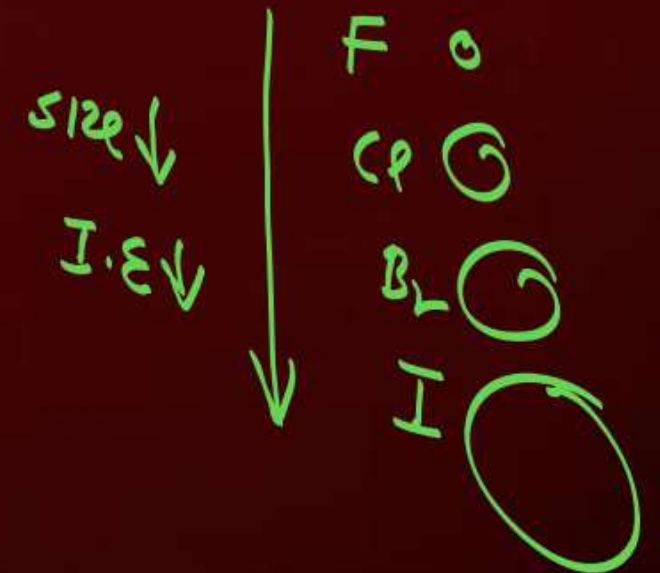
- $\text{IE}_1 < \text{IE}_2 < \text{IE}_3 < \text{IE}_4 \dots$
- **Reason:** removing electrons increases nuclear charge $\rightarrow$ more energy needed



$$\text{IE}_1 < \text{IE}_2 < \text{IE}_3 \dots$$

Variation of IE

- Across a Period: IE $\uparrow$
Nuclear charge $\uparrow$, atomic size $\downarrow \rightarrow$ stronger attraction. (IE $\uparrow$)
- Down a Group: IE $\downarrow$
Atomic size $\uparrow$ (new shells), attraction $\downarrow$.
- **Example:** F > Cl > Br > I



$Z_{eff} \uparrow$ Attraction $\uparrow$ size $\downarrow$ IE $\uparrow$
size $\uparrow$ IE $\downarrow$

Factors Affecting IE

1. **Atomic size:** smaller size $\rightarrow$ higher IE.
2. **Nuclear charge:** greater charge higher IE.
3. **Electronic configuration:** half-filled/fully-filled subshells are more stable $\rightarrow$ higher IE.

★ IE in 2nd Period

General order: $\text{Li} < \boxed{\text{Be}} > \text{B} < \text{C} < \boxed{\text{N}} > \text{O} < \text{F} < \text{Ne}$

p^3 p^6
 $n\tilde{s}n\text{p}^1$

Irregularities:

- Be vs B: $\text{Be} > \text{B}$ (Be has stable $2s^2$).
- N vs O: $\text{N} > \text{O}$ (N has stable half-filled $2p^3$).

2nd.
 $\text{ns}^2 \rightarrow$ fully filled $\rightarrow$ stable $\rightarrow$ IE $\uparrow$
 Be

1s group
 $n\tilde{s}n\text{p}^3$
 N



Electron Gain Enthalpy

- The energy absorbed or released when an electron is added to an isolated gaseous atom in its ground state.
- **Units:** kJ mol^{-1} or eV/atom .
- Δ_{egH} can be negative (energy released) or positive (energy absorbed)



Signs of Δ_{egH}

- **Negative:** Most elements $\rightarrow$ energy released.
- **Positive:** Noble gases, fully/half-filled stable configurations $\rightarrow$ energy absorbed.

Examples:

- Noble gases $\rightarrow \Delta_{\text{egH}}$ positive (stable ns^2np^6).
- Group 2 (ns^2) & Group 15 (ns^2np^3) $\rightarrow$ less negative Δ_{egH} (stability of filled/half-filled orbitals).

Electron Gain Enthalpy

Factors Affecting $\Delta_{eg}H$

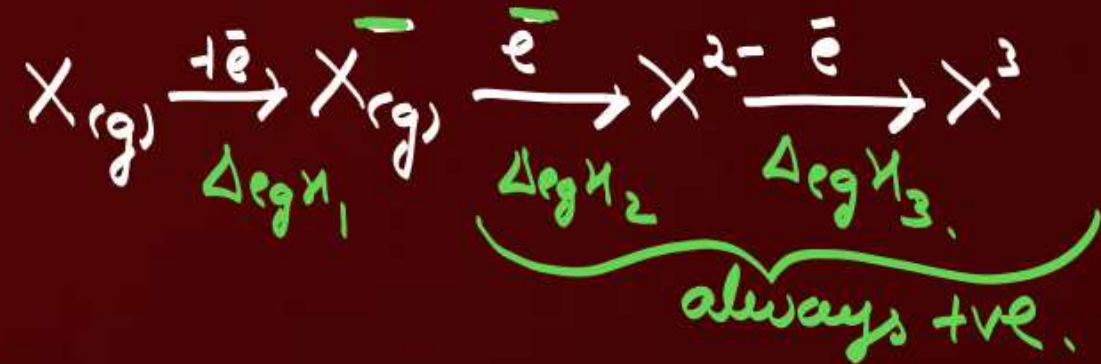
1. **Atomic size:** Smaller size $\rightarrow$ stronger attraction $\rightarrow \Delta_{eg}H$ more negative.
2. **Nuclear charge:** Higher charge $\rightarrow$ stronger attraction $\rightarrow \Delta_{eg}H$ more negative.
3. **Electronic configuration:** Half/fully filled subshells $\rightarrow$ more stable $\rightarrow \Delta_{eg}H$ less negative or positive.

Successive $\Delta_{eg}H$

First $\Delta_{eg}H$: Usually negative (energy released).

Second & further $\Delta_{eg}H$: Always positive

(adding e^- to already negative ion $\rightarrow$ strong repulsion).



Electron Gain Enthalpy

Variation of $\Delta_{eg}H$

SIZE ↓



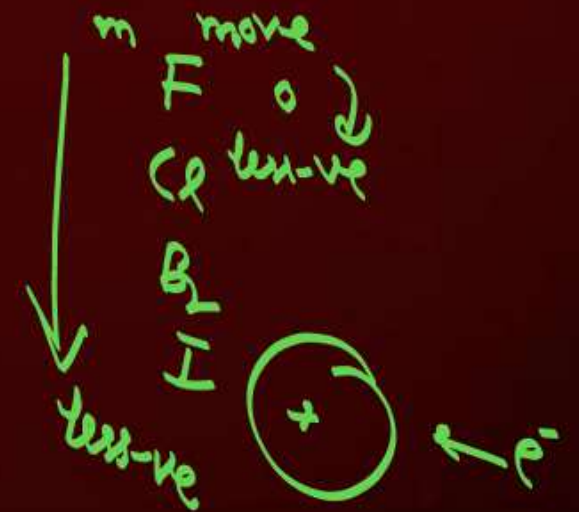
- **Across a period (→):** Atomic number ↑, Z^* ↑, attraction ↑ → $\Delta_{eg}H$ more negative.
- **Down a group (↓):** Atomic size ↑, attraction ↓ → $\Delta_{eg}H$ less negative.

Special Case – Halogens

- **Expected order:** F > Cl > Br > I
- **Experimental order:** Cl > F > Br > I
- **Reason:** Fluorine is very small → high $e^- - e^-$ repulsions → $\Delta_{eg}H$ less negative.
- Chlorine has the most negative $\Delta_{eg}H$.



Inter



Electronegativity



- **Definition:** Ability of an atom to attract shared electrons in a covalent bond.
- **Unit:** Unitless.
- **Scale:** Pauling scale (most used); Fluorine = 4.0 (highest).
- **Most electronegative element:** Fluorine.
- **Least electronegative element:** Cesium (Cs).
- **Variation:**
 - **Across a Period (→):** Increases (nuclear charge ↑, size ↓).
 - **Down a Group (↓):** Decreases (size ↑, attraction ↓).

Valency

- The electrons present in the outermost shell of an atom are called valence electrons and number of valence electrons determine the valency of the atom. The combining capacity of an atom is called its valency.
- In case representative elements ^(s+p) the valency of an atom is generally equal to either the number of valence electrons or equal to $8 - \text{the number of valence electrons}$. But in case of transition and inner-transition elements they show variable valency due to involvement of inner electrons also.



Valency



Variation of valency in a periodic table

(a) Along the period: as we move left to right the valency of the elements first increases from 1 to 4 and then decreases to zero.

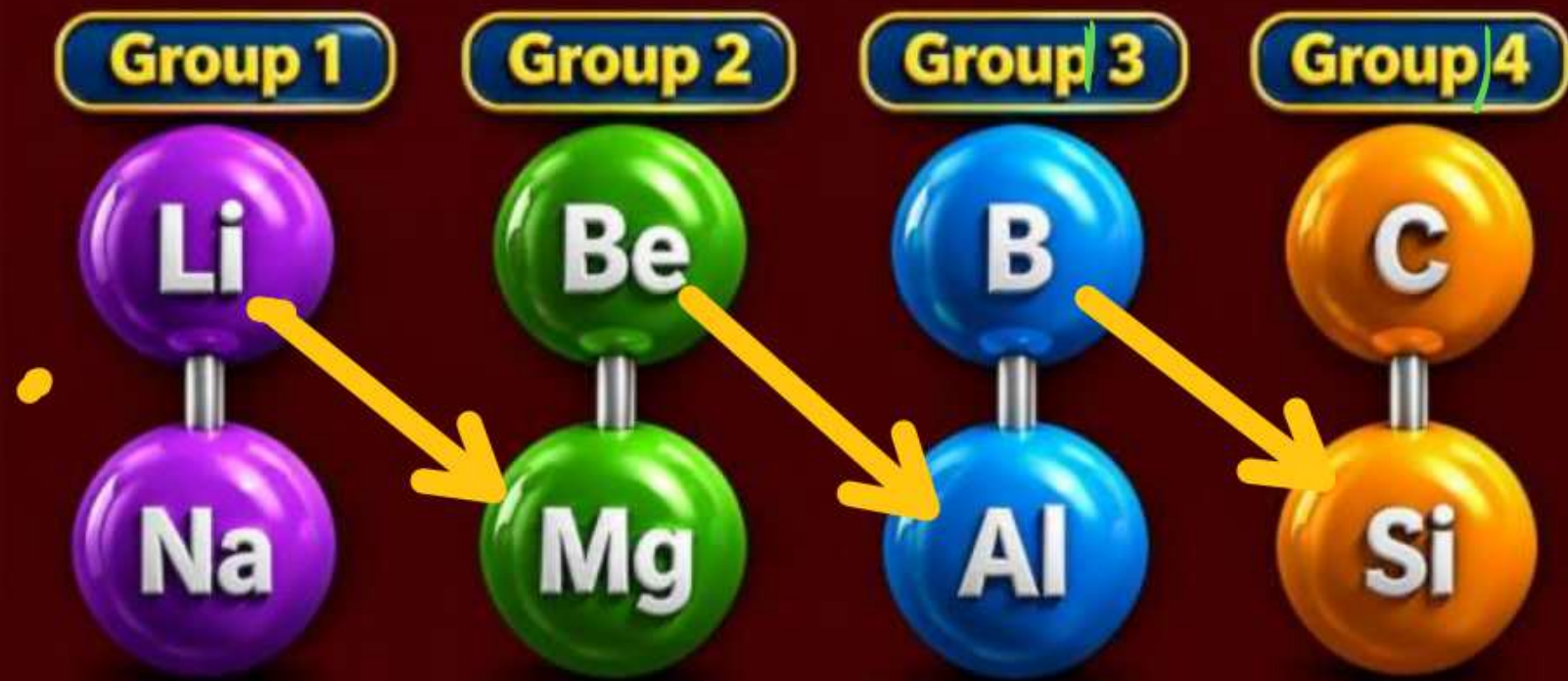
Valence of elements in different groups								
Groups	1	2	13	14	15	16	17	18
No. of valence electrons	1	2	3	4	5	6	7	8
Valence	1	2	3	4	3, 5	2, 6	1, 7	0, 8

(b) Down the group: down the group all the elements have same valency.

For example: First group has 1 Valency.

Diagonal Relationship

- Some elements of second period like lithium, beryllium and boron resemble in their properties with the corresponding element of third period placed diagonally to them.
- This similarity in properties of elements placed diagonally to each other is called the diagonal relationship in the periodic properties. e.g.



QUESTION

Aluminium is diagonally related to:

[2024]

1 Li

2 Be

3 C

4 B

QUESTION



What are *d*-block elements? Write the general electronic configuration of *d*-block elements. **[2025]**

QUESTION

Account for the following:

- (a) Ionization enthalpy of nitrogen is more than oxygen.
- (b) Electron gain enthalpy of fluorine is lesser than chlorine.

[2025]

QUESTION

The diagonal relationship is shown by:

[2024]

- 1** 1st and 2nd period elements
- 2** 1st and 3rd period elements
- 3** 2nd and 3rd period elements
- 4** 3rd and 4th period elements

QUESTION

What are p -block elements? Write the general electronic configuration of p -block elements. [2024]

QUESTION

The most electronegative element in the periodic table is:

[2025]

- 1 Nitrogen
- 2 Oxygen
- 3 Chlorine
- 4 Fluorine

QUESTION



Define ionization enthalpy. What are its units?

[2024]

QUESTION



State modern periodic law. What are the main features of the long form of the periodic table? **[2025]**

QUESTION

What do you understand by representative elements ? Name the groups whose elements are called representative elements. **[2025]**

QUESTION



Which has the smallest size?

[2025]



QUESTION



What do you mean by Periodic properties of elements? Explain. Name any three such properties. **[2025]**

QUESTION



Which of the following ions has the smallest radius?

[2025]

- 1 Cl^-
- 2 S^{2-}
- 3 K^+
- 4 Ca^{2+}

QUESTION

What are transaction elements ? Why are they so called? Write their general electronic configuration. **[2025]**

QUESTION



State and explain

[2025]

- (a) Ionization enthalpy
- (b) Electronegativity

Thank
You